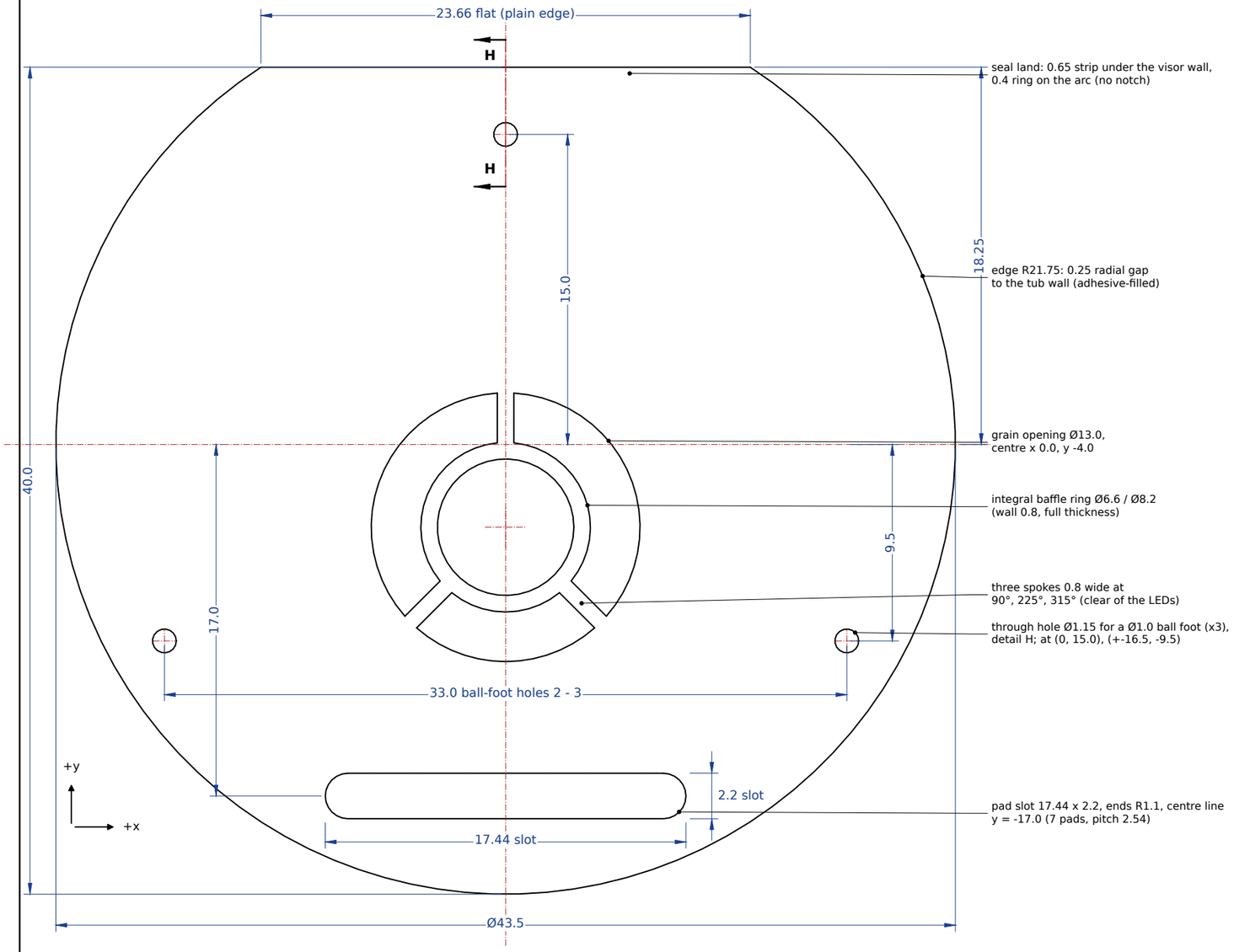
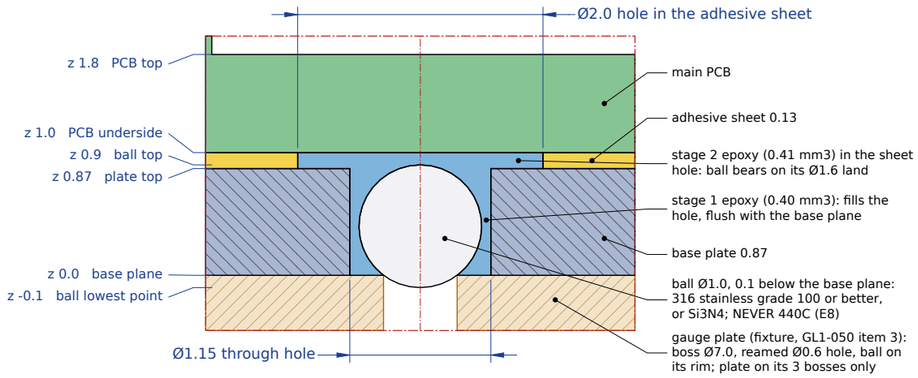
TOP VIEW (PCB side) 4:1



DETAIL H - ball foot 1 as assembled (x = 0), PA12 build 20:1



Feature	Qty	Size	Position (x, y)
Outline	1	R21.75, flat y 18.25	centre (0, 0)
Grain opening	1	Ø13.0 through	(0.0, -4.0)
Baffle ring (integral)	1	Ø6.6 / Ø8.2	concentric with the opening
Spokes	3	0.8 wide	90°, 225°, 315° from +x
Pad slot	1	17.44 x 2.2	centre (0, -17.0)
Ball-foot holes	3	Ø1.15 through	(0, 15.0), (+-16.5, -9.5)
Thickness	-	0.87	z 0 to 0.87

NOTES

1. ALTERNATIVE to GL1-011 (routed FR-4, the primary plate): for an all-printed build and for the EVT comparison of thermal and humidity tilt drift. Plate 0.87 mm with a 0.13 mm adhesive sheet: the PCB underside stays at z = 1.0; seal joint 0.33 mm.
2. Ball feet (Ø1.0: 316 stainless grade 100 or better, or silicon nitride; never 440C, E8): the print cannot hold a 0.1 mm protrusion. Clamp the plate on the three bosses of the gauge plate (reamed Ø0.6 holes), drop the balls in (each seats on the rim of its hole, 0.1 proud), pot them from this side, cure. At lamination the sheet holes are filled with epoxy as well.
3. The baffle ring is part of the plate here (the FR-4 build uses the separate ring GL1-013). The grain opening is potted clear after lamination, in four compartments: surface z 0.05 to 0.10 (target 0.075), never nearer the base plane than z 0.05.
4. The top face is the bonding face for the adhesive sheet and the seal land (r 21.35 to 21.75). Plain edge: no notch.
5. No wall below 0.8 mm, no trapped powder (check 16). Dyed, NOT vapor smoothed (hidden part, three bonding faces).
6. As printed the bureau's tolerance applies (title block). WE sort and finish the plates: thickness 0.83 to 0.92 used (below 0.83 the ball would touch its land through the 0.13 sheet); order about twice the quantity needed. Flatness 0.3 mm free state is OUR target after flattening (between plates under a weight, warm; time and temperature to be found on first articles), not a bureau requirement. We ream the three foot holes to Ø1.15.
7. LEGEND (E10): the content and positions of GL1-011 sheet 2 (maker, GL-1, PRACTICE DEVICE, statement blocks, serial box), LASER-MARKED on the underside (nothing is printed on PA12; no ink height): 1.0 mm clear of the foot holes, the opening and the slot, none inside Ø8.5 round each foot hole. Contrast of a laser mark on dyed black PA12: to be tried on first articles.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE

GL1-012 Base plate, printed PA12 - ALTERNATIVE build

PROJECT / PART

GreenLine GL-1 electronic ball marker - base plate (alternative build, EVT comparison)

MATERIAL / FINISH	PROCESS
PA12 (HP 3D HR PA 12), dyed black	HP Multi Jet Fusion, dyed, NOT vapor smoothed
GENERAL TOLERANCE	SCALE (A3) / UNITS
printed: +-0.3 or +-0.3 % (bureau MJF), as printed	4:1 / 20:1 mm

STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1.2 / D-EVT1	1 of 1 2026-11-13